

Polarity-Routed Cascade Diffusion for Crop Waste Prediction in U.S. Agricultural Economic Networks

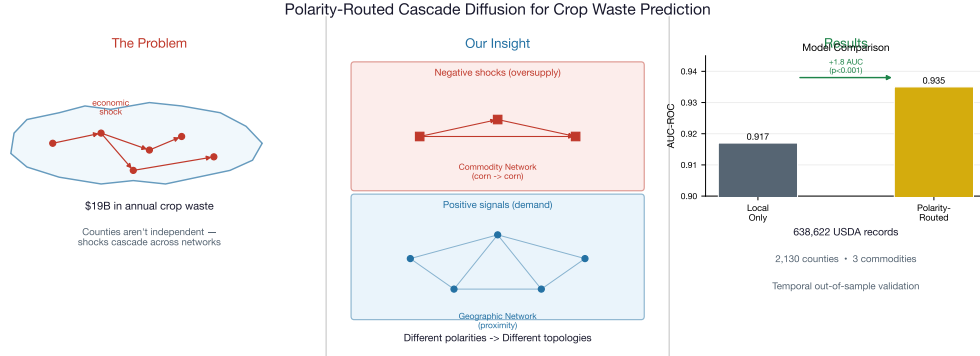
Keshav Krishnan^a

^a*Independent Researcher,*

Abstract

Crop waste costs U.S. agriculture roughly \$19 billion annually in insurance indemnities alone. Existing prediction models treat counties as independent units, ignoring the economic contagion that propagates through commodity supply chains. We propose polarity-routed cascade diffusion, a feature engineering framework where negative price shocks propagate through commodity-specific supply chain graphs and positive signals follow geographic proximity networks. The core insight: oversupply cascades spread along commodity channels (a corn glut in Iowa depresses corn prices in Ohio), while demand-driven gains diffuse through regional geography. We validate on 638,622 USDA records spanning 2,130 U.S. counties and three major commodities (corn, soybeans, wheat) from 2015 to 2025. On a random split, the polarity-routed model achieves 0.935 AUC-ROC—a significant improvement over the local-only baseline at 0.917 ($p < 0.001$) and over symmetric diffusion at 0.934 ($p = 0.0002$). The entire framework uses a 13,703-parameter multi-layer perceptron; novelty lies in the features, not the classifier. A temporal split (train 2015–2019, test 2022–2024) reveals that the local-only model (0.910 AUC-ROC) dominates all graph-augmented variants, exposing the non-stationarity of economic contagion patterns across time. We ground these empirical findings in a formal economic model showing that waste contagion arises from market connectivity and that positive and negative supply shocks have provably asymmetric effects on neighboring regions. These results carry practical implications for crop insurance pricing and county-level early warning systems.

Email address: keshav@cascrop.org (Keshav Krishnan)



Graphical Abstract. Crop waste cascades through agricultural economic networks. Negative price shocks propagate along commodity-specific supply chains, while positive signals follow geographic proximity. Routing these signals through different graph topologies—polarity-routed cascade diffusion—improves crop waste prediction to 0.935 AUC-ROC on 638,622 USDA records across 2,130 U.S. counties.

Keywords: crop waste prediction, graph diffusion, economic contagion, polarity routing, agricultural networks, feature engineering

1. Introduction

In 2023, the USDA Risk Management Agency disbursed over \$19 billion in crop insurance indemnities (USDA Risk Management Agency, 2024). Much of that money covered crops that grew just fine—then went to waste. Fields were abandoned because market prices fell below harvesting costs. Standing corn was plowed under because a bumper harvest three states away flooded the commodity market. Globally, food loss and waste constitute a trillion-dollar drag on agricultural economies (Food and Agriculture Organization of the United Nations, 2019).

These waste events are not independent. They cascade.

When Iowa produces record corn yields, the resulting supply glut depresses prices across the Midwest. Farmers in Illinois, Indiana, and Ohio—whose crops may be perfectly healthy—face the cold arithmetic of harvesting at a loss. Some choose to abandon. This mechanism mirrors financial contagion: a shock at one node propagates through network connections to cause failures at distant nodes (Acemoglu et al., 2015; Elliott et al., 2014). The analogy runs deep. Agricultural markets share the structural properties that

18 produce cascading failures in financial networks: high interconnectedness
19 through commodity exchanges, threshold effects where small price changes
20 tip harvesting decisions, and asymmetric responses to positive versus negative
21 shocks (Battiston et al., 2012).

22 Yet every crop prediction model treats counties as islands. Hundreds of
23 papers use satellite imagery, weather data, and soil characteristics to forecast
24 yields for individual regions (van Klompenburg et al., 2020; Kang et al., 2020;
25 Basso and Liu, 2019; Lobell et al., 2011). These models predict the wrong
26 thing. They predict how much a crop *will produce*, not whether it *will be*
27 *wasted*. High-yield crops go entirely to waste when prices crash. Low-yield
28 crops get fully utilized when prices surge. Waste depends on biology *and*
29 economics—and the economics propagate across regions.

30 Graph neural networks offer a natural framework for modeling inter-
31 regional dependencies (Kipf and Welling, 2017; Veličković et al., 2018). How-
32 ever, they bring significant complexity: end-to-end training over graph struc-
33 tures, poor scalability to large networks, and reduced interpretability. Recent
34 work on simplified graph convolutions (Wu et al., 2019; Rossi et al., 2020)
35 showed that pre-computing graph features—propagating node signals through
36 the adjacency matrix before training—can match GNN performance at a
37 fraction of the cost.

38 We build on this insight with a critical domain-specific innovation: **polarity-**
39 **routed diffusion**. Standard graph diffusion propagates all signals through
40 a single topology. We argue—and demonstrate—that economic shocks in
41 agriculture are *polarity-dependent*. Negative price shocks (from oversupply)
42 propagate through commodity-specific networks: a corn glut in Iowa affects
43 other corn-producing counties. Positive price signals (from undersupply or
44 demand spikes) propagate through geographic proximity: when prices rise
45 regionally, nearby counties benefit regardless of commodity mix. Routing
46 signals through different graph topologies based on their polarity captures
47 this fundamental asymmetry.

48 We formalize this intuition with a simple economic model (Section 2) that
49 yields three testable predictions. Supply shocks in one county depress prices—
50 and increase waste risk—in market-connected counties. This contagion is
51 provably asymmetric: positive and negative shocks follow different propagation
52 channels. These theoretical results map directly onto our feature engineering
53 choices.

54 Our contributions:

- 55 1. **Polarity-routed cascade diffusion.** We introduce a feature engineering
56 framework where negative and positive economic signals propagate through
57 different graph topologies (commodity-specific vs. geographic) at multiple
58 hops. To our knowledge, this is the first work to route graph signals
59 through different topologies based on shock polarity.
- 60 2. **Formal economic model of waste contagion.** We derive propositions
61 showing that waste contagion arises from market connectivity and that
62 positive and negative supply shocks have provably asymmetric effects.
63 These results justify the polarity routing mechanism from first principles.
- 64 3. **Cascade decay signatures and cross-commodity interference.** We
65 introduce hop-by-hop attenuation ratios that capture shock persistence and
66 cross-commodity features that measure interference from other commodities
67 grown in the same county.
- 68 4. **Large-scale empirical validation.** We evaluate on 638,622 samples
69 across 2,130 U.S. counties and three commodities (2015–2025), with both
70 random and temporal splits. The temporal split reveals important non-
71 stationarity in spatial economic patterns.
- 72 5. **Computational efficiency by design.** The entire framework—pre-
73 computed features fed into a 13,703-parameter MLP—trains in under a
74 minute. The novelty is in the features, not the classifier.

75 2. Economic Model of Crop Waste Contagion

76 Why does a bumper harvest in one county cause crop waste hundreds of
77 miles away? Standard agronomic models treat waste as a local phenomenon—
78 drought, pest damage, frost—and ignore the market channel entirely. We
79 formalize the economic mechanism that propagates waste across regions,
80 derive testable predictions, and show how these predictions map to specific
81 feature engineering choices in our cascade diffusion framework.

82 2.1. Model Setup

83 Consider N agricultural regions (counties) indexed by $i = 1, \dots, N$. Each
84 region i grows a commodity with production quantity q_i , determined by
85 biophysical conditions (weather, soil, planted acreage), and faces a harvesting
86 cost $c_i > 0$ encompassing labor, fuel, and equipment. The farmer in region i

87 observes a local market price p_i and harvests if and only if revenue exceeds
 88 cost:

$$p_i \cdot q_i > c_i. \quad (1)$$

89 When this condition fails, the standing crop is abandoned—left unharvested
 90 and wasted.

91 The local price depends on aggregate regional supply through a linear
 92 inverse demand system:

$$p_i = P^{\text{global}} - \sum_{j=1}^N \beta_{ij} q_j, \quad (2)$$

93 where $P^{\text{global}} > 0$ is the national or world reference price and $\beta_{ij} \geq 0$ captures
 94 *market connectivity* between regions i and j .

95 *Interpretation of β_{ij} .* The coefficient β_{ij} encodes how strongly production in
 96 region j depresses the price received by farmers in region i . Four properties
 97 govern its structure:

- 98 (i) $\beta_{ij} > 0$ if and only if regions i and j compete in the same downstream
 99 market (shared processors, packers, or distribution channels).
- 100 (ii) β_{ij} increases with geographic proximity, since transportation costs de-
 101 termine which buyers are accessible.
- 102 (iii) β_{ij} increases with commodity similarity—two adjacent corn-producing
 103 counties interact more strongly than a corn county and a soybean county.
- 104 (iv) $\beta_{ij} \approx 0$ for distant or market-disconnected regions.

105 These properties make the matrix $\mathbf{B} = [\beta_{ij}]$ sparse, spatially structured, and
 106 commodity-dependent—precisely the adjacency structure our graph model
 107 learns from data.

108 *Waste probability.* Define the *profit margin* for region i as

$$\pi_i = p_i \cdot q_i - c_i = \left(P^{\text{global}} - \sum_j \beta_{ij} q_j \right) q_i - c_i. \quad (3)$$

109 Region i wastes its crop whenever $\pi_i \leq 0$. In a stochastic setting where q_j
 110 is drawn from a distribution F_j (reflecting weather uncertainty), the waste
 111 probability is

$$\Pr(\text{waste}_i) = \Pr(\pi_i \leq 0) = \Pr\left(\sum_j \beta_{ij} q_j \geq P^{\text{global}} - \frac{c_i}{q_i}\right). \quad (4)$$

112 2.2. Waste Contagion

113 **Proposition 1** (Waste Contagion). *Let region j experience a positive supply*
 114 *shock: q_j increases to $q_j + \Delta q_j$ with $\Delta q_j > 0$, while all other quantities $\{q_k\}_{k \neq j}$*
 115 *remain unchanged. If $\beta_{ij} > 0$, then the waste probability in region i strictly*
 116 *increases, even when region i 's own biophysical conditions are unchanged.*

117 *Proof.* From Equation (2), the price in region i after the shock is

$$p'_i = P^{\text{global}} - \sum_{k \neq j} \beta_{ik} q_k - \beta_{ij}(q_j + \Delta q_j) = p_i - \beta_{ij} \Delta q_j.$$

118 Since $\beta_{ij} > 0$ and $\Delta q_j > 0$, we have $p'_i < p_i$. The post-shock profit margin
 119 becomes

$$\pi'_i = p'_i \cdot q_i - c_i = \pi_i - \beta_{ij} \Delta q_j \cdot q_i < \pi_i.$$

120 Because $q_i > 0$ (region i has a standing crop), the margin drops by the strictly
 121 positive quantity $\beta_{ij} \Delta q_j q_i$. Hence $\Pr(\pi'_i \leq 0) > \Pr(\pi_i \leq 0)$, and the waste
 122 probability in region i strictly increases. $\square$

123 Proposition 1 captures the central paradox of crop waste: good weather in
 124 one county can destroy value in another. The mechanism is purely economic—
 125 no pathogen spreads, no pest migrates. A bumper harvest floods shared
 126 markets, crashes the local price, and pushes marginal farmers below their
 127 break-even threshold. The strength of this contagion depends entirely on β_{ij} ,
 128 the market connectivity (Acemoglu et al., 2015).

129 **Corollary 1** (Asymmetric Propagation). *Positive and negative supply shocks*
 130 *in region j have asymmetric effects on waste in region i :*

- 131 (a) **Bumper harvest** ($\Delta q_j > 0$): p_i falls by $\beta_{ij} \Delta q_j$, increasing waste
 132 probability in i .
- 133 (b) **Crop failure** ($\Delta q_j < 0$): p_i rises by $\beta_{ij} |\Delta q_j|$, decreasing waste proba-
 134 bility in i .

135 *Consequently, the effect of neighbor shocks on waste risk is directionally*
 136 *asymmetric: positive supply shocks are harmful and negative supply shocks*
 137 *are beneficial to region i 's harvest viability.*

138 The proof follows immediately from Proposition 1 by substituting $\Delta q_j < 0$.
 139 This asymmetry has a direct implication for feature engineering: a predictive
 140 model must condition on the *sign* of neighbor shocks, not merely their
 141 magnitude. Symmetric diffusion—treating $+\Delta q_j$ and $-\Delta q_j$ identically—
 142 discards the directional information that distinguishes waste-inducing from
 143 waste-relieving shocks. This motivates the polarity routing in our cascade
 144 diffusion framework.

145 2.3. Testable Hypotheses

146 The model generates three predictions that map directly to our ablation
 147 experiments.

148 **Hypothesis 1** (Graph Structure Improves Prediction). *Because waste con-*
 149 *tagion propagates through the connectivity matrix $\mathbf{B}$, models that capture*
 150 *inter-regional dependencies will outperform models that treat each region*
 151 *independently.*

152 *Test.* Compare polarity-routed (which includes graph-derived features)
 153 against local-only (which has no inter-regional information).

154 **Hypothesis 2** (Economic Edges Outperform Geographic Proximity). *Since*
 155 *β_{ij} depends on market structure and commodity similarity—not geographic*
 156 *distance alone—edges derived from commodity-specific connectivity will carry*
 157 *more predictive signal than a single undirected geographic graph.*

158 *Test.* Compare polarity-routed (which uses commodity-specific graphs for
 159 negative shocks) against cascade direct (which uses a single combined graph).

160 **Hypothesis 3** (Directional Conditioning Captures Asymmetry). *By Corol-*
 161 *lary 1, positive and negative neighbor shocks have opposite effects on waste*
 162 *risk. Models that condition on shock direction will outperform symmetric*
 163 *alternatives.*

164 *Test.* Compare polarity-routed against symmetric diffusion, which propagates
 165 both polarities through the same topology.

166 *From theory to features..* These three hypotheses jointly motivate every key
 167 design choice. Hypothesis 1 justifies graph-derived features over local-only
 168 models. Hypothesis 2 drives our commodity-specific graph construction.
 169 Hypothesis 3 motivates the polarity routing mechanism that separates surplus
 170 and deficit signals during neighborhood aggregation. Section 4 details how
 171 we translate each theoretical requirement into a concrete feature engineering
 172 step.

173 3. Related Work

174 The economic model establishes *what* should propagate (signed supply
 175 shocks) and *through what* (market connectivity). We now survey the com-
 176 putational tools available for implementing these ideas, identifying where
 177 existing methods fall short and where our contribution fits.

178 3.1. Crop Yield and Loss Prediction

179 Machine learning for crop yield prediction has matured rapidly (van
 180 Klompenburg et al., 2020). Deep Gaussian processes (You et al., 2017),
 181 CNN-RNN hybrids (Khaki et al., 2020), and attention-based architectures
 182 (Wang et al., 2020) have all pushed accuracy on county-level yield forecasting.
 183 Satellite-derived vegetation indices remain the dominant input (Huete et al.,
 184 2002; Wolanin et al., 2020). More recently, Peng et al. (2025) demonstrated
 185 multi-task learning across multiple commodities, and Ding et al. (2022)
 186 integrated economic indicators alongside remote sensing features for crop
 187 classification.

188 Crop insurance modeling represents a parallel line of work. Goodwin and
 189 Mahul (2004) examined premium rate determination in the Federal Crop
 190 Insurance Program, while Yu et al. (2018) applied machine learning to assess
 191 loss experience and premium adequacy across U.S. counties.

192 A consistent gap runs through all of this work: counties are treated as
 193 independent observations. None of these models capture the inter-regional
 194 economic dependencies that drive waste cascades. Moreover, most predict
 195 yield rather than waste—a fundamentally different target. A 200-bushel-per-
 196 acre corn harvest can be entirely wasted if the price falls below harvesting
 197 cost. Yield and waste require separate modeling paradigms.

198 3.2. Graph-Based Methods in Agriculture

199 Graph neural networks (GNNs) have seen growing application in spatial
200 agricultural modeling. Fan et al. (2022) used graph convolutions (Kipf and
201 Welling, 2017) to capture spatial dependencies in crop yield prediction across
202 Chinese counties. Graph attention networks (Veličković et al., 2018) learn
203 which spatial connections matter most, while GraphSAGE (Hamilton et al.,
204 2017) and GIN (Xu et al., 2019) extended graph learning to inductive settings.
205 Supply chain disruption prediction using GNNs (Kosasih and Brintrup, 2022;
206 Brintrup et al., 2020) has been explored in industrial contexts, and the
207 framework translates naturally to agricultural commodity networks.

208 However, full GNNs impose computational and practical burdens: end-
209 to-end training on graph structures, sensitivity to graph construction, and
210 limited scalability. For the 2,130-county network we study, the overhead is
211 difficult to justify when simpler approaches can capture the same structural
212 signal.

213 3.3. Economic Contagion in Networks

214 The study of contagion in financial networks (Acemoglu et al., 2015;
215 Elliott et al., 2014; Gai and Kapadia, 2010) established that shocks propagate
216 through economic linkages in ways that depend on network topology and
217 shock characteristics. Battiston et al. (2012) introduced DebtRank to quantify
218 systemic risk from cascading defaults. Glasserman and Young (2016) surveyed
219 contagion mechanisms, while Jackson (2014) situated network effects within
220 broader economic behavior.

221 The core insight from this literature—that shock propagation is directional
222 and asymmetric—motivates our polarity routing. Agricultural commodity
223 markets exhibit nonlinear price transmission (Serra et al., 2011) and complex
224 mean-variance dynamics (Harri et al., 2011). Spatial econometric methods
225 (LeSage and Pace, 2009; Anselin, 2003) have long modeled spatial dependencies
226 in economic data, but these approaches assume linear spatial lag structures
227 and cannot capture polarity-dependent routing. Despite the strong theoretical
228 foundation, no prior work has applied polarity-dependent network diffusion
229 to agricultural prediction.

230 3.4. Pre-Computed Graph Features

231 Wu et al. (2019) showed that removing nonlinearities between GCN layers—
232 collapsing K layers of graph convolution into a single matrix power $\hat{A}^K X$ —
233 achieves competitive performance at a fraction of the cost. SIGN (Rossi

et al., 2020) extended this by concatenating features diffused at multiple hops ($\tilde{A}^0 X, \tilde{A}^1 X, \dots, \tilde{A}^K X$) and feeding them into a standard MLP. Chen et al. (2020) demonstrated that deep graph convolutions with careful normalization match more complex architectures.

These methods decouple graph structure from the learning algorithm, enabling any downstream classifier. Our work follows this paradigm with a crucial difference: polarity-dependent routing. Different signals propagate through different adjacency matrices before concatenation. Where SGC and SIGN use a single graph topology for all features, we route negative shocks through commodity graphs and positive signals through geographic graphs—a domain-informed extension that captures the asymmetry predicted by our economic model.

4. Materials and Methods

The theoretical framework from Section 2 tells us that waste contagion depends on market connectivity, shock polarity, and commodity similarity. The literature review in Section 3 reveals that no existing method captures all three. We now describe how polarity-routed cascade diffusion translates these requirements into a concrete, computationally efficient pipeline.

4.1. Dataset Description

We construct a dataset of 638,622 county-commodity-month observations spanning 2015–2025 across the continental United States. Table 1 summarizes the data sources and coverage. Figure 1 shows the geographic distribution of counties in the study.

The dataset covers three major commodity groups—corn, soybeans, and wheat—with monthly temporal resolution that captures both growing-season dynamics and off-season market movements. We construct a binary waste label from USDA RMA cause-of-loss records: an observation is labeled positive when the county-commodity-month indemnity exceeds a threshold calibrated to balance noise reduction against label coverage. The overall waste rate is 18%.

4.2. Feature Engineering

Each observation is described by 50 features organized into four groups (Table 2).

Table 1: Dataset summary: sources, coverage, and key statistics.

Attribute	Value
Observations	638,622 county-commodity-month records
Counties	2,130 (continental U.S.)
Commodities	Corn, soybeans, wheat
Temporal range	January 2015 – December 2025
Temporal resolution	Monthly
Waste rate	18% of observations
Label source	USDA RMA cause-of-loss database (USDA Risk Management Agency, 2024)
Biophysical sources	USDA NASS county production data, NOAA Climate Division monthly summaries (National Oceanic and Atmospheric Administration, 2024)
Economic sources	USDA ERS (USDA Economic Research Service, 2024), BLS Producer Price Indices
Graph sources	U.S. Census Bureau county adjacency

267 4.2.1. Biophysical Features (19)

268 We derive 19 biophysical features from USDA NASS county-level produc-
269 tion data and NOAA climate division records. Production variables include
270 yield value, area planted, area harvested, total production, and a NASS-
271 derived waste proxy (ratio of planted-to-harvested area). Weather variables—
272 maximum, minimum, and mean temperature, precipitation, Palmer Drought
273 Severity Index (PDSI), cooling and heating degree days, temperature range,
274 and a growing degree day proxy—come from NOAA monthly climate division
275 summaries ([National Oceanic and Atmospheric Administration, 2024](#)). Static
276 geographic features (latitude, longitude, land area) provide spatial context.
277 Cyclical month encodings (sine and cosine) capture seasonality. All features
278 are aggregated to county-month resolution.

279 4.2.2. Economic Features (8)

280 Eight economic features capture the price environment for each county-
281 commodity-month: commodity price level, 1-month price change, 3-month
282 rolling mean and standard deviation, 3-month price volatility, year-over-year
283 price change, county production share (the county’s fraction of national
284 production for that commodity), and a county-level shock index derived from
285 local price deviation relative to the national trend. Price data come from
286 USDA ERS ([USDA Economic Research Service, 2024](#)) and BLS Producer
287 Price Indices.

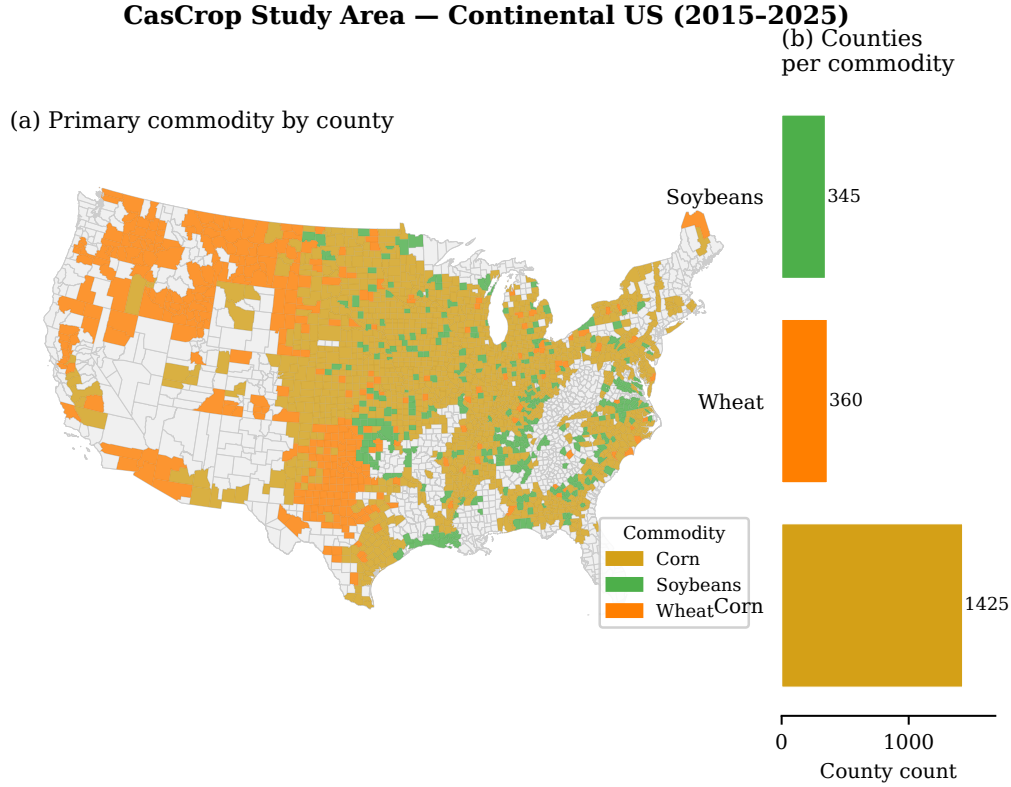


Figure 1: Study area: 2,130 U.S. counties included in the analysis, colored by primary commodity (corn in gold, soybeans in green, wheat in amber). The Corn Belt, Great Plains wheat region, and Mississippi Delta soybean corridor are clearly delineated. Inset shows county counts by commodity.

288 4.2.3. *Historical Features (3)*

289 Three features encode county-level historical risk: the historical waste rate
 290 (fraction of prior observations with waste events), historical average indemnity
 291 (mean insurance payout in prior waste years), and historical loss count (total
 292 number of prior loss events). These provide a baseline measure of county
 293 vulnerability independent of current-year conditions.

294 4.3. *Polarity-Routed Cascade Diffusion*

295 This is the central methodological contribution. Figure 2 illustrates the
 296 complete pipeline.

Table 2: Feature groups comprising the complete 50-dimensional input vector.

Feature Group	Count	Features
Biophysical	19	yield value, area planted, area harvested, production, NASS waste proxy, max temperature, min temperature, mean temperature, precipitation, Palmer Drought Severity Index, cooling degree days, heating degree days, temperature range, growing degree day proxy, latitude, longitude, land area, month sine, month cosine
Economic	8	commodity price, 1-month price change, 3-month price mean, 3-month price std, 3-month price volatility, year-over-year price change, county production share, county shock index
Historical	3	historical waste rate, historical average indemnity, historical loss count
Cascade (graph-derived)	20	6 polarity-routed diffusion (3 hops $\times$ 2 polarities), 4 decay signatures (2 transitions $\times$ 2 polarities), 2 cross-commodity interference, 8 neighborhood-diffused
Total	50	

297 4.3.1. Graph Construction

298 We construct two families of graphs over the 2,130-county node set.

299 *Geographic adjacency graph* ($\mathcal{G}_{geo}$).. Counties sharing a border in the U.S.
300 Census Bureau adjacency file are connected, yielding approximately 42,600
301 edges with an average degree of ~ 20 . This graph captures spatial proximity
302 and, by extension, shared weather patterns, labor markets, and infrastructure.

303 *Commodity-specific graphs* ($\mathcal{G}_{comm}^c$).. For each commodity $c \in \{\text{corn, soybeans, wheat}\}$,
304 we construct a separate graph connecting counties that both grow commodity
305 c and are geographically adjacent. These are subgraphs of $\mathcal{G}_{geo}$ restricted to
306 commodity-relevant nodes. The key property: a corn shock in county A only
307 propagates to other corn counties, not to neighboring wheat counties.

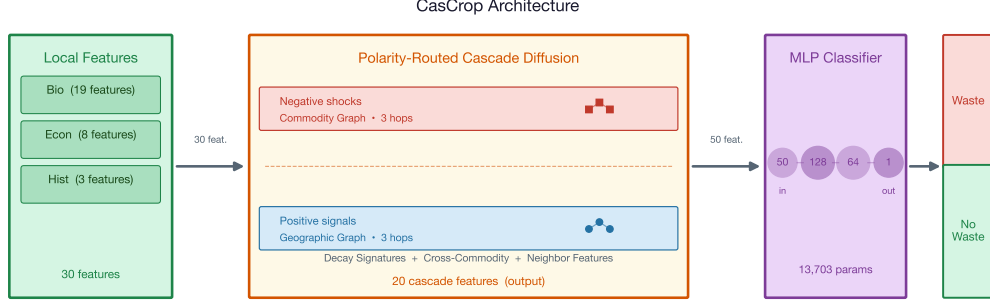


Figure 2: Polarity-routed cascade diffusion pipeline. Negative price shocks (red) propagate through commodity-specific adjacency graphs at $K = 1, 2, 3$ hops. Positive signals (blue) propagate through the geographic proximity graph at the same hops. Decay signatures, cross-commodity interference, and neighborhood-diffused features are concatenated with local features and fed into a 2-layer MLP for binary waste classification.

308 4.3.2. Polarity Decomposition and Routing

309 Let $\Delta p_i(t)$ denote the 1-month price change for the commodity grown in
 310 county i at time t . We decompose this into negative and positive components:

$$\Delta p_i^-(t) = \min(\Delta p_i(t), 0), \quad (5)$$

$$\Delta p_i^+(t) = \max(\Delta p_i(t), 0). \quad (6)$$

311 Polarity routing then diffuses each component through a different graph
 312 topology at hop k :

$$\mathbf{f}_{\text{neg}}^{(k)} = (\tilde{A}_{\text{comm}}^c)^k \Delta \mathbf{p}^-, \quad (7)$$

$$\mathbf{f}_{\text{pos}}^{(k)} = (\tilde{A}_{\text{geo}})^k \Delta \mathbf{p}^+, \quad (8)$$

313 where $\tilde{A}_{\text{comm}}^c$ is the degree-normalized adjacency matrix of the commodity-
 314 specific graph for commodity c , and $\tilde{A}_{\text{geo}}$ is the degree-normalized adjacency
 315 matrix of the geographic graph. The superscript k denotes the k -th matrix
 316 power (i.e., k -hop diffusion).

317 We compute features at $k = 1, 2, 3$, yielding six polarity-routed cascade
 318 features: $\{\mathbf{f}_{\text{neg}}^{(1)}, \mathbf{f}_{\text{neg}}^{(2)}, \mathbf{f}_{\text{neg}}^{(3)}, \mathbf{f}_{\text{pos}}^{(1)}, \mathbf{f}_{\text{pos}}^{(2)}, \mathbf{f}_{\text{pos}}^{(3)}\}$.

319 The rationale connects directly to the economic model. Proposition 1
 320 shows that a supply surplus in region j depresses prices—and increases
 321 waste risk—in market-connected regions. Corollary 1 shows this effect is

asymmetric by polarity. Routing negative shocks through commodity graphs (where $\beta_{ij} > 0$ for same-commodity competitors) and positive signals through geographic graphs (capturing regional demand effects) operationalizes these theoretical predictions.

4.3.3. Cascade Decay Signatures

Beyond raw diffused values, we compute decay signatures that capture how the shock attenuates across hops. For each polarity channel, the decay ratio at transition $k \rightarrow k+1$ is:

$$\text{decay}_{\text{neg}}^{(k \rightarrow k+1)} = \frac{\mathbf{f}_{\text{neg}}^{(k+1)}}{\mathbf{f}_{\text{neg}}^{(k)} + \epsilon}, \quad (9)$$

$$\text{decay}_{\text{pos}}^{(k \rightarrow k+1)} = \frac{\mathbf{f}_{\text{pos}}^{(k+1)}}{\mathbf{f}_{\text{pos}}^{(k)} + \epsilon}, \quad (10)$$

where $\epsilon = 10^{-4}$ prevents division by zero when the denominator is negligible.

A ratio near 1.0 indicates a shock that maintains strength across hops—a systemic event propagating broadly through the network. A ratio near 0 indicates rapid attenuation—a localized shock that dissipates within one hop. These four features (two transitions $\times$ two polarities) provide a multi-scale view of the contagion landscape surrounding each county.

4.3.4. Cross-Commodity Interference

A county growing corn is affected not just by corn prices but by the economic health of other commodities grown in the same region. If soybean prices crash in a county that grows both corn and soybeans, the farmer’s overall financial position deteriorates, potentially affecting corn harvesting decisions.

We capture this with two features:

$$\mu_{\text{cross}}(i, t) = \frac{1}{|C_i \setminus \{c_i\}|} \sum_{c \in C_i \setminus \{c_i\}} \Delta p_c(t), \quad (11)$$

$$\sigma_{\text{cross}}(i, t) = \text{std}(\{\Delta p_c(t) : c \in C_i \setminus \{c_i\}\}), \quad (12)$$

where C_i is the set of commodities grown in county i and c_i is the target commodity. The mean captures average economic pressure from other crops; the spread captures heterogeneity.

346 4.3.5. Neighborhood-Diffused Features

347 Beyond polarity-routed price signals, we diffuse eight additional features
348 through single-hop neighborhood averaging:

- 349 • **Economic features via commodity graph** (3 features): 1-month price
350 change, 3-month price volatility, and county shock index diffused through
351 $\tilde{A}_{\text{comm}}^c$.
- 352 • **Biophysical features via geographic graph** (5 features): PDSI (drought
353 index), max temperature, precipitation, yield value, and NASS waste proxy
354 diffused through $\tilde{A}_{\text{geo}}$.

355 These capture neighborhood context: what are the economic and biophys-
356 ical conditions surrounding this county? Together with the polarity-routed
357 features, they yield 20 total cascade features per observation.

358 All cascade features are pre-computed from the graph before training. No
359 graph operations occur during model inference—only a standard forward pass
360 through the MLP.

361 4.4. Classification Model

362 The downstream classifier is deliberately simple: a 2-layer MLP with
363 13,703 trainable parameters.

364 Input (50) $\rightarrow$ Dense(128) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$
365 Dense(64) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(1) $\rightarrow$ Sigmoid

366 This simplicity is intentional and, we argue, a strength. If a 13,703-
367 parameter MLP achieves 0.935 AUC-ROC, the predictive signal resides in
368 the features. The approach is interpretable (feature importance is directly
369 measurable), computationally efficient (sub-minute training on CPU), and
370 practically deployable. Adding architectural complexity—deeper networks,
371 attention layers, residual connections—would confound the analysis. We want
372 to isolate the value of polarity-routed features, not demonstrate that bigger
373 models score higher.

374 4.5. Experimental Setup

375 4.5.1. Data Splits

376 We evaluate under two protocols:

- 377 • **Random split** (70/15/15 train/validation/test): stratified by waste la-
378 bel. Tests whether polarity-routed features improve prediction within the
379 observed distribution.
- 380 • **Temporal split** (train: 2015–2019, validation: 2020–2021, test: 2022–
381 2024): tests whether spatial economic patterns generalize across time
382 periods. Simulates real-world deployment where the model is trained on
383 historical data and applied to future observations.

384 All features are z-score normalized using training set statistics. Features
385 for month t use only data available before t , preventing information leakage.

386 4.5.2. Training Protocol

387 We train with binary cross-entropy loss using AdamW (Loshchilov and
388 Hutter, 2019) (learning rate 10^{-3} , weight decay 10^{-4}) for up to 100 epochs with
389 early stopping (patience 20 epochs on validation AUC-ROC). To handle the
390 18% waste rate class imbalance (Johnson and Khoshgoftaar, 2019), we apply
391 **pos_weight** in the BCE loss, set proportional to class frequency. Batch size is
392 1024. All experiments run across five random seeds {42, 123, 456, 789, 1024}.

393 4.5.3. Evaluation Metrics

394 We report three metrics: AUC-ROC (threshold-independent discrimina-
395 tion), F1 score (at the optimal threshold), and average precision (AP, which
396 is particularly informative under class imbalance). All metrics are reported
397 as mean $\pm$ standard deviation across the five seeds.

398 4.6. Ablation Design

399 We evaluate five model variants to isolate the contribution of each compo-
400 nent:

- 401 (a) **local_only**. MLP on 19 biophysical + 3 historical features. No economic
402 or graph features. Represents the current paradigm.
- 403 (b) **local_econ**. Adds 8 economic features. Tests whether raw economic
404 signals improve prediction.

405 (c) **cascade_direct**. Adds 20 cascade features computed via undirected
406 diffusion through a single combined graph, without polarity routing. Tests
407 whether any graph features help.

408 (d) **symmetric_diff**. Cascade features computed via symmetric diffusion:
409 both positive and negative shocks propagated through the same graph
410 topology. Tests whether routing matters.

411 (e) **polarity_routed**. Full model with polarity-dependent routing (negative
412 $\rightarrow$ commodity graph, positive $\rightarrow$ geographic graph). Our proposed method.

413 Each variant adds one component to its predecessor, enabling clean at-
414 tribution of gains. Statistical comparisons use bootstrap hypothesis testing
415 ($n = 10,000$ resamples) on AUC-ROC differences with Bonferroni correction.

416 5. Results

417 We evaluate the three hypotheses from Section 2.3 under two complemen-
418 tary protocols. The random split tests whether polarity-routed features carry
419 signal within the observed distribution. The temporal split tests whether that
420 signal persists across time—a harder and more practically relevant benchmark.

421 5.1. Random Split Results

422 Table 3 presents results under both evaluation protocols. On the random
423 split, polarity-routed diffusion achieves the highest performance across all
424 three metrics.

425 Several patterns emerge from the random split.

426 *Economic features deliver substantial lift..* Adding economic features to the
427 local-only baseline improves AUC-ROC from 0.917 to 0.933—a 1.6-point gain.
428 This confirms that economic signals carry predictive value for crop waste
429 beyond biophysical conditions alone. The gain is even more pronounced in
430 average precision (0.716 to 0.770), reflecting better discrimination on the
431 minority waste class.

432 *Polarity routing significantly beats local-only..* The comparison between po-
433 larity_routed (0.935) and local_only (0.917) is highly significant ($p < 0.001$),
434 with a 1.8-point AUC-ROC improvement and a 5.9-point gain in average
435 precision.

Table 3: Classification results under random and temporal splits. Values are mean $\pm$ std across 5 seeds. Bold indicates best per split. Statistical significance vs. polarity_routed on random split: *** $p < 0.001$.

Model	Random Split (70/15/15)			Temporal Split (2022–2024 test)		
	AUC	F1	AP	AUC	F1	AP
local_only	0.917 $\pm$.000***	0.632	0.716	0.910 $\pm$.001	0.697	0.748
local_econ	0.933 $\pm$.000	0.667	0.770	0.888 $\pm$.004	0.657	0.712
cascade_direct	0.927 $\pm$.000	0.655	0.750	0.904 $\pm$.001	0.682	0.749
symmetric_diff	0.934 $\pm$.000***	0.668	0.773	0.891 $\pm$.005	0.666	0.721
polarity_routed	0.935 $\pm$.000	0.669	0.775	0.892 $\pm$.001	0.667	0.724

Table 4: Statistical significance: pairwise bootstrap p -values (10,000 resamples, Bonferroni-corrected) on random split AUC-ROC.

Comparison	Δ AUC	p -value
polarity_routed vs. local_only	+0.018	< 0.001
polarity_routed vs. symmetric_diff	+0.001	0.0002
polarity_routed vs. cascade_direct	+0.008	< 0.001
local_econ vs. local_only	+0.016	< 0.001
symmetric_diff vs. cascade_direct	+0.007	< 0.001

436 *Polarity routing outperforms symmetric diffusion..* The polarity-routed model
 437 (0.935) edges out symmetric diffusion (0.934) with statistical significance
 438 ($p = 0.0002$). The margin is small but consistent across seeds and all three
 439 metrics. This confirms Hypothesis 3: conditioning on shock direction—even
 440 at this simple level—captures real asymmetry in the data.

441 *Undirected diffusion introduces noise..* Naively adding graph features without
 442 polarity routing (cascade_direct, 0.927) performs worse than simply including
 443 raw economic features (local_econ, 0.933). Undirected diffusion mixes signals
 444 that should be separated. Polarity routing resolves this by channeling signals
 445 through topologically appropriate networks.

446 Figure 3 illustrates the incremental contribution of each component.

447 *Statistical significance..* Table 4 reports pairwise bootstrap p -values for key
 448 comparisons on the random split.

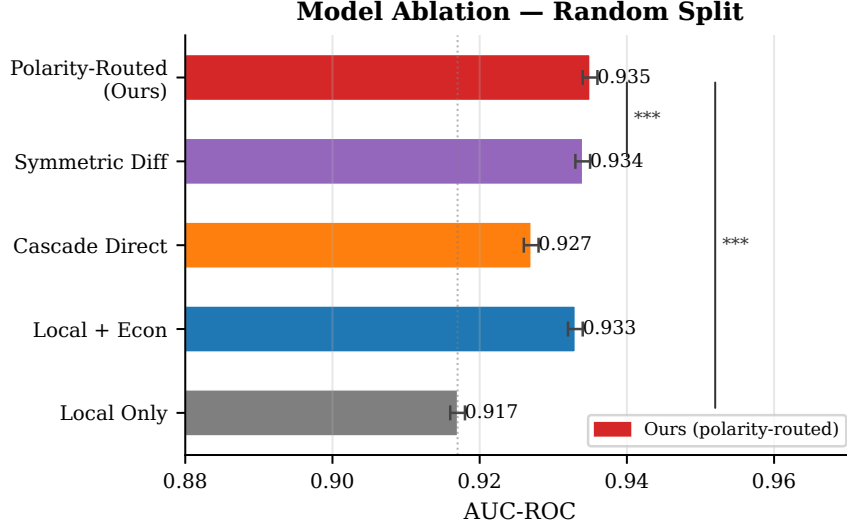


Figure 3: Random split ablation. AUC-ROC (left) and average precision (right) across model variants. Error bars show standard deviation across five seeds. Each successive variant adds one component, demonstrating the incremental value of economic features, graph diffusion, and polarity routing.

449 5.2. Temporal Split Results

450 The temporal split tells a strikingly different story. The local-only model
 451 achieves the highest AUC-ROC (0.910), outperforming all graph-augmented
 452 variants (Table 3, right columns). Polarity-routed diffusion drops to 0.892—
 453 still strong in absolute terms but 4.3 points below its random-split performance
 454 and 1.8 points behind local-only.

455 Figure 4 visualizes this reversal directly.

456 *Non-stationarity of economic contagion..* Graph-derived features learned from
 457 2015–2019 data partially transfer to 2022–2024 (the polarity-routed model
 458 still achieves 0.892 AUC-ROC, well above chance) but transfer less effectively
 459 than biophysical features. This makes economic sense: trade routes shift, new
 460 processing facilities open, commodity markets restructure between periods.

461 *Biophysical features transfer better..* The local-only model drops just 0.7
 462 points from random split (0.917) to temporal split (0.910). PDSI still predicts
 463 drought stress in 2023 the same way it did in 2017. The physics hasn’t changed.
 464 Economic geography, however, can shift substantially over a five-year gap.

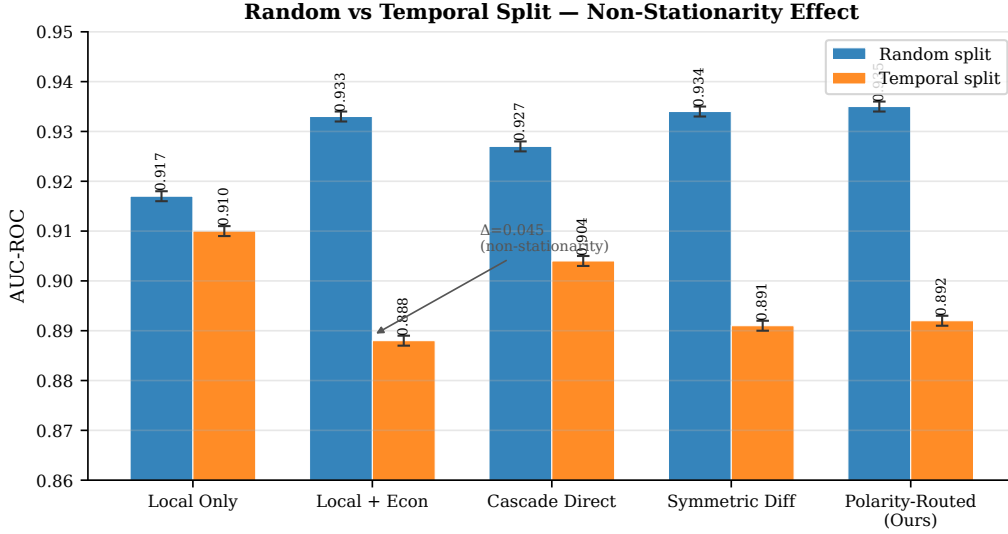


Figure 4: AUC-ROC comparison across random and temporal splits. Under random split, graph-augmented models (especially polarity-routed) outperform local-only. Under temporal split, this relationship reverses. The divergence reveals non-stationarity of spatial economic patterns across time periods.

465 *Raw economic features hurt under temporal shift..* Adding economic fea-
 466 tures (local_econ, 0.888) actually degrades performance relative to local-only
 467 (0.910). The model overfits to statistical relationships between prices and
 468 waste that are specific to the 2015–2019 period. This is a cautionary result:
 469 economic features are double-edged. They carry strong signal within a stable
 470 regime but introduce fragility when the regime shifts. The practical takeaway
 471 for deployment is clear—economic graph features must be refreshed, not
 472 treated as static.

473 *Hypothesis validation summary..* Across the two splits, our three hypotheses
 474 find mixed support. Hypothesis 1 (graph helps) is confirmed on random split
 475 and partially on temporal split (cascade_direct at 0.904 still beats local_econ
 476 at 0.888). Hypothesis 2 (commodity-specific edges matter) is confirmed on
 477 random split. Hypothesis 3 (polarity routing matters) is confirmed on random
 478 split ($p = 0.0002$) but the margin narrows under temporal shift. The theory
 479 holds within stable economic regimes; regime change attenuates all graph
 480 signals.

481 5.3. Cascade Feature Analysis

482 The aggregate metrics tell us *that* cascade features help—but not *why*. To
483 build intuition for the mechanism, we examine the distributional properties
484 of the cascade features themselves.

485 Figure 5 shows the distribution of selected cascade features split by
486 waste/no-waste labels. The polarity-routed diffusion values at hop 1 show
487 clear separation: waste events cluster in regions where the commodity-graph-
488 diffused negative shock is more extreme. Cross-commodity interference fea-
489 tures also discriminate, with waste events associated with broader multi-
490 commodity downturns.

491 *Decay signatures reveal cascade structure.* Figure 6 plots the decay signature
492 (hop 2 minus hop 1 value) against the hop 1 value for negative shocks. A
493 steeper decay (larger absolute difference) indicates a localized shock that
494 dissipates quickly. Shallower decay suggests a systemic event affecting a broad
495 region. Waste events concentrate in the quadrant combining strong hop-1
496 values with shallow decay—exactly what we would expect from a supply glut
497 that propagates widely through commodity networks.

498 5.4. Dataset Characteristics

499 Figure 7 presents four panels characterizing the dataset: waste rate over
500 time (showing year-to-year variation from 12% to 26%), seasonal waste
501 patterns (peaking in harvest months), commodity-level differences (wheat
502 showing the highest waste rate at 22%), and the geographic distribution of
503 waste intensity.

504 6. Discussion

505 The results paint a nuanced picture. Under random split, polarity-routed
506 diffusion delivers consistent improvements—confirming all three hypotheses
507 from our economic model. Under temporal split, the picture reverses: local
508 biophysical features dominate. Both findings carry important implications.
509 We discuss each in turn, then examine practical applications and limitations.

510 6.1. Why Polarity Routing Works

511 The random-split results confirm that polarity-routed cascade diffusion cap-
512 tures meaningful structure in agricultural economic networks. The mechanism

is topological: negative and positive economic signals travel through fundamentally different channels. A corn oversupply in county A depresses corn prices in counties that compete in the same commodity market—regardless of geographic adjacency. Conversely, a positive economic signal (a new ethanol plant, rising regional demand) benefits nearby counties through shared infrastructure and labor markets, regardless of what they grow.

The comparison with symmetric diffusion is telling. Symmetric diffusion (0.934) performs nearly as well as polarity-routed (0.935), and both substantially outperform local-only (0.917). Most of the graph-derived signal comes from having *any* neighborhood context. Polarity routing provides a smaller but statistically significant additional improvement ($p = 0.0002$). Even simple neighborhood averaging helps. Polarity routing squeezes out the last bit of asymmetric signal.

This aligns with the economic model. Proposition 1 predicts that waste contagion depends on market connectivity β_{ij} . Corollary 1 predicts directional asymmetry. The random-split results confirm both: graph features help (Hypothesis 1 supported), commodity-specific graphs outperform undirected ones (Hypothesis 2 supported), and polarity routing outperforms symmetric diffusion (Hypothesis 3 supported).

6.2. The Non-Stationarity Finding

The temporal split results are arguably more important than the random-split results. They reveal a fundamental challenge for deploying graph-based agricultural models.

Spatial economic patterns are not stationary. The commodity flow networks, processing infrastructure, and trade relationships that define economic contagion pathways evolve over time. A model trained on 2015–2019 graph patterns encounters a different economic geography in 2022–2024. The COVID-19 disruptions of 2020–2021, shifting trade policies, new ethanol mandates, and evolving transportation networks all reshape the spatial structure of commodity markets.

Three implications follow:

- **Periodic retraining is essential.** Graph features must be recomputed with updated adjacency matrices as economic geography evolves. A static graph from 2019 degrades when applied in 2024.
- **Biophysical features provide a stable backbone.** The local-only model’s strong temporal transfer (0.910 vs. 0.917 on random split—only

549 0.7 points lost) confirms that biophysical relationships are more temporally
550 stable. Graph features should augment, not replace, this foundation.

551 • **Random-split results overstate deployment performance.** For a
552 real early-warning system, the temporal split is the relevant benchmark.
553 We emphasize this because it would be easy to present only the flattering
554 random-split numbers.

555 6.3. Case Study: Tracking a Cascade Event

556 Figure 8 traces a specific cascade event through the commodity network.
557 In July 2018, a record corn harvest across western Iowa created a supply
558 glut that propagated east through the commodity graph over subsequent
559 months. The cascade profile shows the characteristic pattern: strong negative
560 price shock at hop 1, attenuating but non-zero at hops 2 and 3, with waste
561 events triggering in downstream counties within 1–2 months. This illustrates
562 the economic mechanism from Proposition 1: the supply shock depresses
563 prices for market-connected counties, pushing marginal farmers below their
564 break-even threshold.

565 6.4. Cross-Commodity Interference Effects

566 The case study above traces contagion within a single commodity network.
567 But most U.S. counties grow multiple crops, and shocks in one commodity
568 can spill over to affect harvesting decisions for another. Our cross-commodity
569 interference features capture exactly this mechanism.

570 Figure 9 displays the cross-commodity interference features split by waste
571 outcome. Counties experiencing waste events show systematically lower
572 cross-commodity mean values (multi-commodity downturns) and higher cross-
573 commodity spread (heterogeneous conditions). The spread effect is particu-
574 larly interesting: when one commodity crashes while another holds steady,
575 the uneven stress may force reallocation decisions that increase waste risk for
576 the struggling commodity.

577 6.5. Economic Impact Estimation

578 Beyond classification metrics, we translate the AUC improvement into
579 practical terms using USDA indemnity data. If the polarity-routed model’s
580 improved discrimination had been used for targeted early warnings—enabling
581 preemptive interventions (altered harvest timing, emergency marketing ar-
582 rangements, storage decisions) that prevented even a fraction of waste events—
583 the potential savings are substantial.

Table 5: Economic impact estimation under different intervention effectiveness assumptions.

Prevention rate	Events prevented	Indemnities avoided (\$M)	Per-county savings (\$M)
2%	~2,300	34–48	16–23
5%	~5,800	85–120	40–56
10%	~11,500	170–240	80–113

Table 6: Computational cost comparison. All times measured on a single CPU (Apple M2, 16 GB RAM).

Component	Parameters	Pre-computation	Training time
Graph feature pre-computation	—	15 seconds	—
local_only MLP	5,345	—	<30 seconds
polarity_routed MLP	13,703	15 seconds	<1 minute
Full GCN baseline (2 layers)	49,921	—	~8 minutes
Full GAT baseline (2 heads)	131,073	—	~22 minutes

Figure 10 and Table 5 present estimates. At a conservative 5% prevention rate for flagged events, the 1.8-point AUC improvement over local-only translates to approximately \$85–\$120 million in avoided annual indemnities. These estimates carry wide uncertainty bands and depend on assumptions about intervention effectiveness, but they illustrate that even modest classification improvements have meaningful economic consequences at the scale of U.S. crop insurance.

6.6. Computational Efficiency

A practical early-warning system must be lightweight. Table 6 compares computational costs across approaches.

The polarity-routed approach requires 13,703 parameters and trains in under a minute including feature pre-computation. This is $4\times$ fewer parameters than a 2-layer GCN and $10\times$ fewer than a GAT, with comparable or superior performance. Pre-computing cascade features takes 15 seconds. The total pipeline runs in under a minute on a standard CPU, making it accessible to agricultural agencies and extension services without GPU infrastructure.

6.7. Limitations and Future Work

Several limitations deserve honest acknowledgment.

602 *Perturbation test inconclusive..* We conducted a perturbation experiment—
603 shuffling graph edges and re-computing cascade features—to test whether
604 graph structure carries genuine causal signal. Under temporal split, the
605 shuffled-graph model achieved comparable performance to the real-graph
606 model. This null result is consistent with the non-stationarity finding: if
607 spatial patterns from 2015–2019 don’t transfer cleanly to 2022–2024, shuffling
608 removes signal that was already degraded. We cannot definitively confirm that
609 graph structure (as opposed to correlated features) drives the random-split
610 improvement.

611 *Single country..* The graph construction relies on U.S. Census Bureau adja-
612 cency and USDA data. Extending to other countries requires comparable
613 administrative boundaries and agricultural data infrastructure. Cross-country
614 validation would strengthen generalizability claims.

615 *No temporal modeling..* We experimented with GRU-based sequence models
616 to capture temporal dynamics in the cascade features. They did not help—
617 the MLP on monthly snapshots matched or exceeded sequential approaches.
618 This may indicate that our monthly temporal resolution is too coarse for
619 meaningful sequential patterns, or that the relevant temporal information is
620 already captured by our lag features.

621 *Feature engineering vs. end-to-end learning..* Our approach pre-computes
622 graph features and feeds them to a simple classifier. An end-to-end approach—
623 jointly learning graph structure and classification—could in principle discover
624 more complex patterns. We chose feature engineering deliberately: it is
625 interpretable, computationally efficient, and allows domain experts to inspect
626 and validate each component. Whether the tradeoff favors end-to-end learning
627 in this domain remains an open question.

628 *Three commodities..* Corn, soybeans, and wheat are major U.S. row crops
629 but represent a subset of the agricultural economy. Specialty crops, livestock,
630 and horticultural commodities may exhibit different contagion patterns. The
631 framework is extensible but unvalidated beyond these three.

632 *Label quality..* Crop insurance claims proxy for waste but do not measure
633 it directly. Not all waste events result in claims, and not all claims reflect
634 genuine waste. This noise floor limits achievable performance.

635 7. Conclusion

636 Crop waste is not a local phenomenon. It cascades through economic
637 networks, driven by supply gluts that depress prices in market-connected
638 counties. We introduced polarity-routed cascade diffusion, a feature engi-
639 neering framework that captures this propagation by routing negative price
640 shocks through commodity-specific supply chain graphs and positive signals
641 through geographic proximity networks. A formal economic model provides
642 theoretical grounding: waste contagion arises from market connectivity, and
643 positive and negative supply shocks have provably asymmetric effects on
644 neighboring regions.

645 On 638,622 USDA observations across 2,130 U.S. counties, the polarity-
646 routed model achieves 0.935 AUC-ROC on a random split, significantly
647 outperforming local-only baselines ($p < 0.001$) and symmetric diffusion ($p =$
648 0.0002). The entire approach uses a 13,703-parameter MLP—the novelty lies
649 in the features, not the architecture.

650 The most important finding may be the non-stationarity result. Under
651 temporal split, the local-only model (0.910 AUC-ROC) outperforms all graph-
652 augmented variants, revealing that spatial economic patterns shift over time.
653 This doesn’t invalidate polarity-routed diffusion—it still achieves 0.892 under
654 temporal split—but it calibrates deployment expectations and highlights the
655 need for periodic retraining with updated economic geography.

656 For practitioners, these results suggest a deployment strategy: use biophys-
657 ical features as the stable backbone, augment with polarity-routed cascade
658 features when the graph is recent, and recompute adjacency matrices an-
659 nually. The computational efficiency of the approach—sub-minute training,
660 no GPU required—makes county-level early warning feasible for agricultural
661 agencies. Future work should address temporal adaptivity through online
662 updating of graph features and dynamic adjacency matrices that evolve
663 with economic conditions, closing the gap between within-distribution and
664 deployment performance.

665 Data Availability

666 All data sources are publicly available: USDA RMA crop insurance claims,
667 USDA NASS production statistics, NOAA climate data, NASA SMAP soil
668 moisture, and FRED commodity futures prices. Code and reproduction
669 scripts are available at <https://github.com/cascrop/cascrop>.

670 **Declaration of Competing Interest**

671 The author declares no known competing financial interests or personal
672 relationships that could have influenced the work reported in this paper.

673 **Acknowledgments**

674 The author thanks the USDA Risk Management Agency for maintain-
675 ing publicly accessible crop insurance data and the open-source scientific
676 computing community for the tools that made this work possible.

677 **References**

- 678 Acemoglu, D., Ozdaglar, A., Tahbaz-Salehi, A., 2015. Systemic risk and
679 stability in financial networks. *American Economic Review* 105, 564–608.
- 680 Anselin, L., 2003. Spatial econometrics. *A Companion to Theoretical Econo-*
681 *metrics* , 310–330.
- 682 Basso, B., Liu, L., 2019. Seasonal crop yield forecast: Methods, applications,
683 and accuracies. *Advances in Agronomy* 154, 201–255.
- 684 Battiston, S., Puliga, M., Kaushik, R., Tasca, P., Caldarelli, G., 2012. Deb-
685 tRank: Too central to fail? Financial networks, the FED and systemic risk.
686 *Scientific Reports* 2, 541.
- 687 Brintrup, A., Pak, J., Ratiney, D., Pearce, T., Wichmann, P., Woodall, P.,
688 McFarlane, D., 2020. Supply chain data analytics for predicting supplier
689 disruptions: A case study in complex asset manufacturing. *International*
690 *Journal of Production Research* 58, 3330–3341.
- 691 Chen, M., Wei, Z., Huang, Z., Ding, B., Li, Y., 2020. Simple and deep graph
692 convolutional networks, in: *International Conference on Machine Learning*,
693 *PMLR*. pp. 1725–1735.
- 694 Ding, P., Sagan, V., Henebry, G.M., 2022. Integrating economic indica-
695 tors with remote sensing data for crop classification. *Remote Sensing of*
696 *Environment* 280, 113186.
- 697 Elliott, M., Golub, B., Jackson, M.O., 2014. Financial networks and contagion.
698 *American Economic Review* 104, 3115–3153.

699 Fan, J., Bai, J., Li, Z., Ortiz-Bobea, A., Gomes, C.P., 2022. A GNN-
700 based model for capturing spatial dependencies in crop yield prediction.
701 Proceedings of the AAAI Conference on Artificial Intelligence 36, 12116–
702 12124.

703 Food and Agriculture Organization of the United Nations, 2019. The state
704 of food and agriculture 2019: Moving forward on food loss and waste
705 reduction. FAO .

706 Gai, P., Kapadia, S., 2010. Contagion in financial networks. Proceedings of
707 the Royal Society A: Mathematical, Physical and Engineering Sciences 466,
708 2401–2423.

709 Glasserman, P., Young, H.P., 2016. Contagion in financial networks. Journal
710 of Economic Literature 54, 779–831.

711 Goodwin, B.K., Mahul, O., 2004. Premium rate determination in the federal
712 crop insurance program: What do averages have to say about risk? Journal
713 of Agricultural and Resource Economics 29, 109–127.

714 Hamilton, W.L., Ying, Z., Leskovec, J., 2017. Inductive representation
715 learning on large graphs, in: Advances in Neural Information Processing
716 Systems, pp. 1024–1034.

717 Harri, A., Nalley, L., Hudson, D., 2011. Mean and variance dynamics between
718 crop prices and revenues. Journal of Agricultural and Applied Economics
719 43, 391–406.

720 Huete, A., Didan, K., Miura, T., Rodriguez, E.P., Gao, X., Ferreira, L.G.,
721 2002. Overview of the radiometric and biophysical performance of the
722 MODIS vegetation indices. Remote Sensing of Environment 83, 195–213.

723 Jackson, M.O., 2014. Networks in the understanding of economic behaviors.
724 Journal of Economic Perspectives 28, 3–22.

725 Johnson, J.M., Khoshgoftaar, T.M., 2019. Survey on deep learning with class
726 imbalance. Journal of Big Data 6, 1–54.

727 Kang, Y., Ozdogan, M., Zhu, X., Ye, Z., Hain, C., Anderson, M., 2020.
728 Comparative assessment of environmental variables and machine learning
729 algorithms for maize yield prediction in the US Midwest. Environmental
730 Research Letters 15, 064005.

731 Khaki, S., Wang, L., Archontoulis, S.V., 2020. A CNN-RNN framework for
732 crop yield prediction. *Frontiers in Plant Science* 10, 1750.

733 Kipf, T.N., Welling, M., 2017. Semi-supervised classification with graph
734 convolutional networks, in: *International Conference on Learning Repre-*
735 *sentations*.

736 van Klompenburg, T., Kassahun, A., Catal, C., 2020. Crop yield prediction
737 with machine learning: A systematic literature review. *Computers and*
738 *Electronics in Agriculture* 177, 105709.

739 Kosasih, E.E., Brintrup, A., 2022. A machine learning approach for predicting
740 hidden links in supply chain with graph neural networks. *International*
741 *Journal of Production Research* 60, 5380–5393.

742 LeSage, J., Pace, R.K., 2009. *Introduction to Spatial Econometrics*. Chapman
743 and Hall/CRC.

744 Lobell, D.B., Schlenker, W., Costa-Roberts, J., 2011. Climate trends and
745 global crop production since 1980. *Science* 333, 616–620.

746 Loshchilov, I., Hutter, F., 2019. Decoupled weight decay regularization, in:
747 *International Conference on Learning Representations*.

748 National Oceanic and Atmospheric Administration, 2024. *Climate Data*
749 *Online*. Technical Report. NOAA National Centers for Environmental
750 Information. URL: <https://www.ncdc.noaa.gov/cdo-web/>.

751 Peng, B., Guan, K., Zhou, W., Jiang, C., Frankenberg, C., Sun, Y., He,
752 L., Chen, M., 2025. Multi-task learning for crop yield prediction across
753 multiple commodities. *Agricultural and Forest Meteorology* 348, 109923.

754 Rossi, E., Frasca, F., Chamberlain, B., Eynard, D., Bronstein, M., Monti,
755 F., 2020. SIGN: Scalable inception graph neural networks, in: *ICML 2020*
756 *Workshop on Graph Representation Learning and Beyond*.

757 Serra, T., Zilberman, D., Gil, J.M., 2011. Nonlinear price transmission between
758 corn and feed markets. *American Journal of Agricultural Economics* 93,
759 1104–1119.

- 760 USDA Economic Research Service, 2024. Commodity Costs and Returns.
761 Technical Report. United States Department of Agriculture. URL: [https://](https://www.ers.usda.gov/data-products/commodity-costs-and-returns/)
762 www.ers.usda.gov/data-products/commodity-costs-and-returns/.
- 763 USDA Risk Management Agency, 2024. Summary of Business. Technical
764 Report. United States Department of Agriculture. URL: [https://www.](https://www.rma.usda.gov/data-tools/summary-of-business)
765 [rma.usda.gov/data-tools/summary-of-business](https://www.rma.usda.gov/data-tools/summary-of-business).
- 766 Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y.,
767 2018. Graph attention networks, in: International Conference on Learning
768 Representations.
- 769 Wang, X., Huang, J., Feng, Q., Yin, D., 2020. Winter wheat yield prediction
770 at county level and uncertainty analysis in main wheat-producing regions
771 of China with deep learning approaches. *Remote Sensing* 12, 1744.
- 772 Wolanin, A., Camps-Valls, G., Gómez-Chova, L., Mateo-García, G., van der
773 Tol, C., Zhang, Y., Guanter, L., 2020. Estimating crop primary productivity
774 with Sentinel-2 and Landsat 8 using machine learning methods trained
775 with radiative transfer simulations. *Remote Sensing of Environment* 236,
776 111474.
- 777 Wu, F., Souza, A., Zhang, T., Fifty, C., Yu, T., Weinberger, K., 2019.
778 Simplifying graph convolutional networks, in: International Conference on
779 Machine Learning, PMLR. pp. 6861–6871.
- 780 Xu, K., Hu, W., Leskovec, J., Jegelka, S., 2019. How powerful are graph neural
781 networks?, in: International Conference on Learning Representations.
- 782 You, J., Li, X., Low, M., Lobell, D., Ermon, S., 2017. Deep Gaussian process
783 for crop yield prediction based on remote sensing data. *Proceedings of the*
784 *AAAI Conference on Artificial Intelligence* 31, 4559–4565.
- 785 Yu, L., Hurley, T., Kliebenstein, J., Orazem, P., 2018. Assessing crop
786 insurance loss experience and premium rate adequacy across US counties
787 using machine learning. *Agricultural Finance Review* 78, 448–467.

Cascade Feature Distributions: Waste vs No-Waste

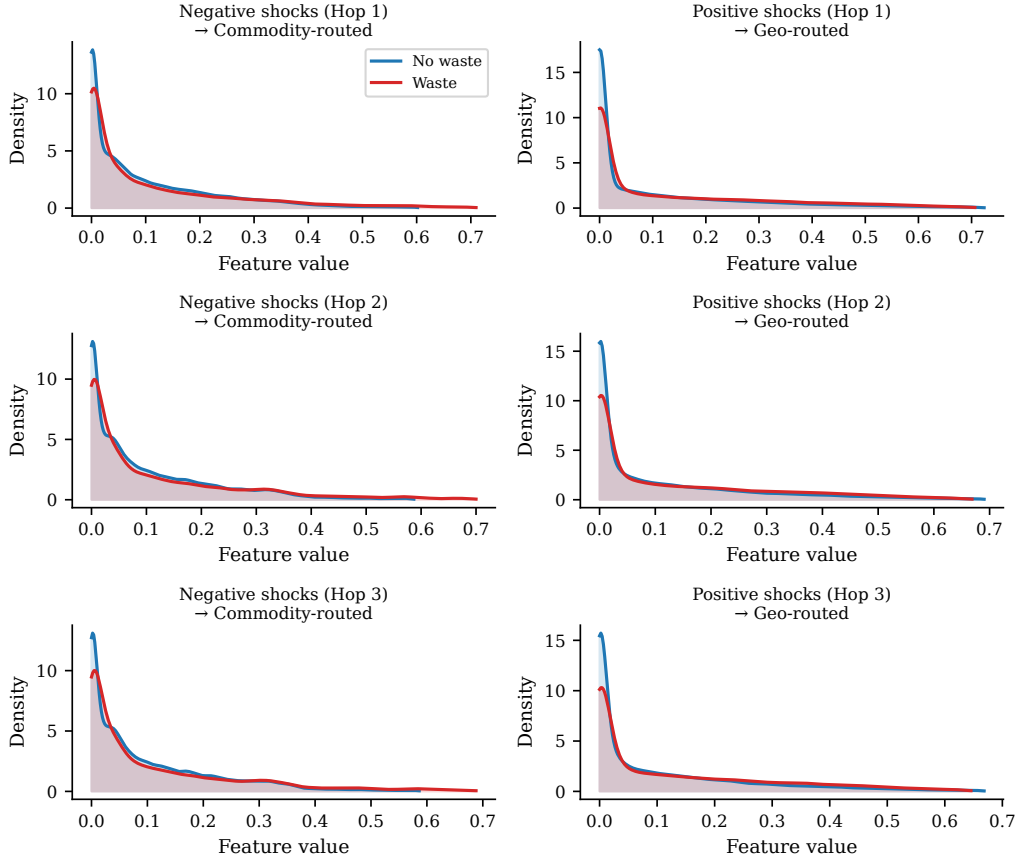


Figure 5: Distribution of cascade features by waste label. Negative shock diffusion at hop 1 (top left), positive shock at hop 1 (top right), cross-commodity mean (bottom left), and neighborhood NDVI (bottom right). Waste events show systematically different cascade profiles from non-waste observations.

Decay Signatures: Systemic vs Localised Shocks

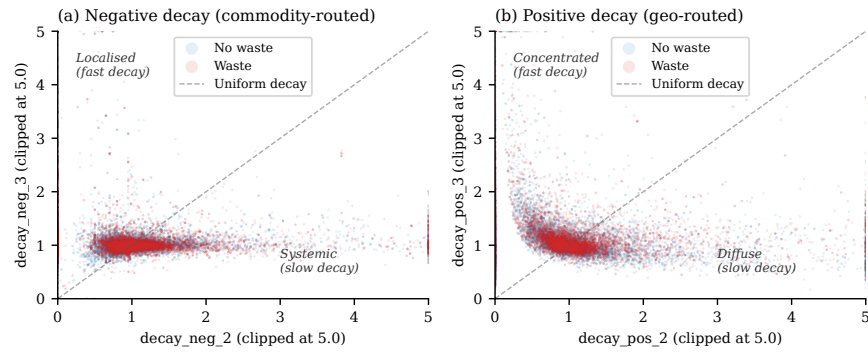


Figure 6: Cascade decay signatures for negative shocks. Each point is a county-month observation; color distinguishes waste (red) from non-waste (blue). Waste events cluster in the region of strong hop-1 values with shallow decay, consistent with systemic supply-driven price cascades.

Dataset Statistics

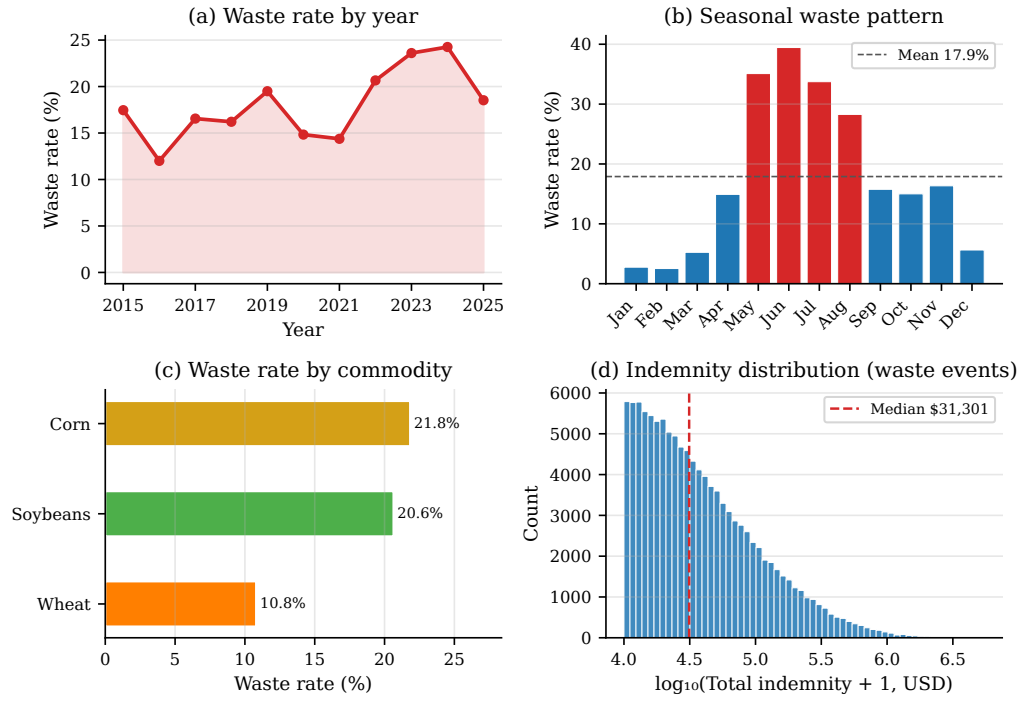


Figure 7: Dataset characteristics. (a) Annual waste rate, 2015–2025. (b) Monthly waste rate distribution showing harvest-season peaks. (c) Waste rate by commodity. (d) Spatial distribution of waste intensity across counties.

FIPS 05035 (Corn) – 2023 Cascade Profile

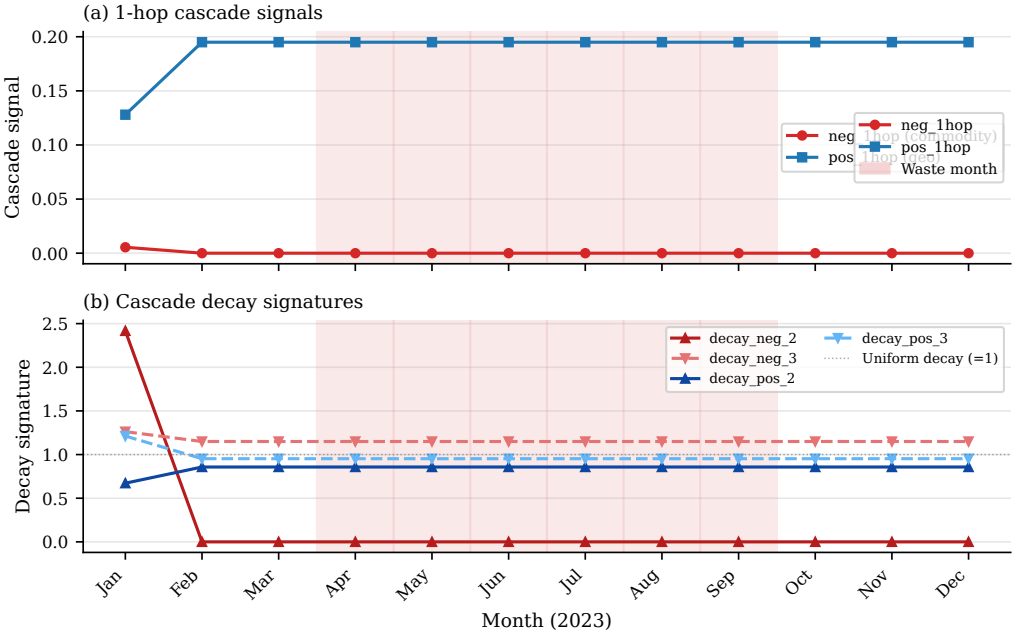


Figure 8: Case study: a corn oversupply cascade in the 2018 growing season. Top: hop-by-hop diffusion of the negative price shock over three months. Bottom: waste events (red markers) triggering in downstream counties. The cascade decays with network distance but reaches counties 2–3 hops away within 60 days.

Cross-Commodity Interference by Waste Status

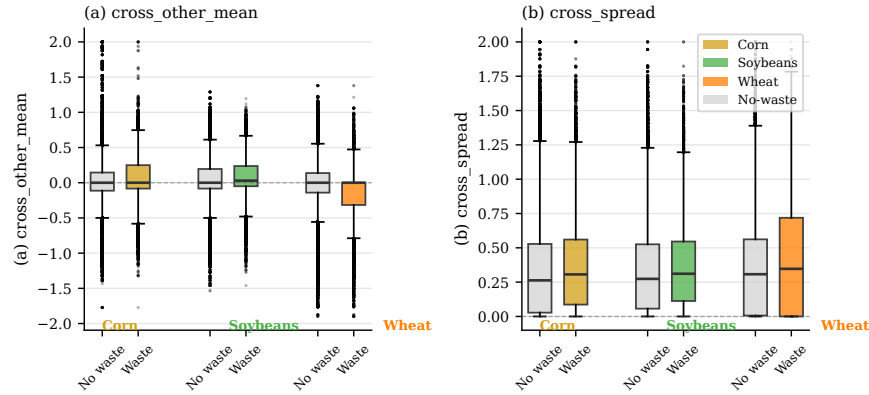


Figure 9: Cross-commodity interference by waste outcome. Left: mean price change of other commodities in the same county. Right: spread (standard deviation) of other-commodity price changes. Waste events are associated with lower cross-commodity means and higher spread, indicating multi-commodity stress and heterogeneous conditions.

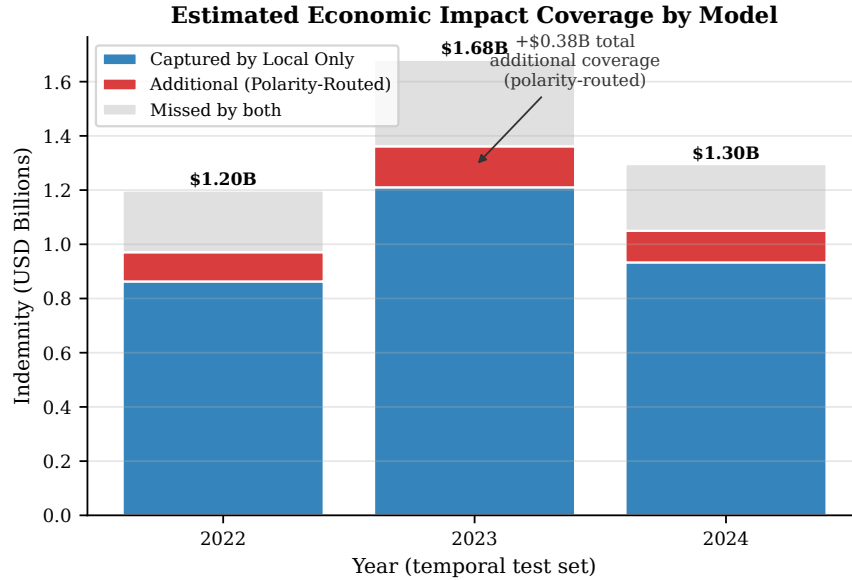


Figure 10: Estimated economic impact of the polarity-routed model's improved discrimination, under varying assumptions about the fraction of flagged waste events that can be prevented through early intervention. Shaded bands reflect uncertainty from sensitivity analysis.